Klorvatran Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-763/B

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Deviations observed during the campaign were investigated and closed with no impact to product quality. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. No changes to the approved analytical procedures are proposed in this submission. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles.

Deviations observed during the campaign were investigated and closed with no impact to product quality. The control strategy links critical quality attributes to critical process parameters identified during development. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. No changes to the approved analytical procedures are proposed in this submission. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. The control strategy links critical quality attributes to critical process parameters identified during development. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

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Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. The control strategy links critical quality attributes to critical process parameters identified during development.

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Statistical evaluation of batch release data demonstrates process capability well within specification limits. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. The control strategy links critical quality attributes to critical process parameters identified during development. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles.

P.4 Control of Excipients

For the commercial formulation F-763/B, Pearlitol 200SD (Mannitol, diluent) is used.

Justification of Material Grade

Pearlitol 200SD was selected as the commercial grade of mannitol on the basis of robust tablet hardness across the compression force design space.

The excipient control strategy was developed in accordance with IPEC Excipient Qualification Guide, ICH Q8(R2) and ICH Q3D.

Commercial Control Strategy — Excipients

Component	Function	Grade per current specification
Mannitol	diluent	Pearlitol 200SD
Reference		SPEC-EX-3807 Rev 08

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Cleaning verification swab results were below the health-based exposure limits derived for the product. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. The control strategy links critical quality attributes to critical process parameters identified during development. Deviations observed during the campaign were investigated and closed with no impact to product quality. No changes to the approved analytical procedures are proposed in this submission.

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

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